

# Cerebral Blood Flow Differences Between IGE and Heathy Controls and their Relationship with Mood

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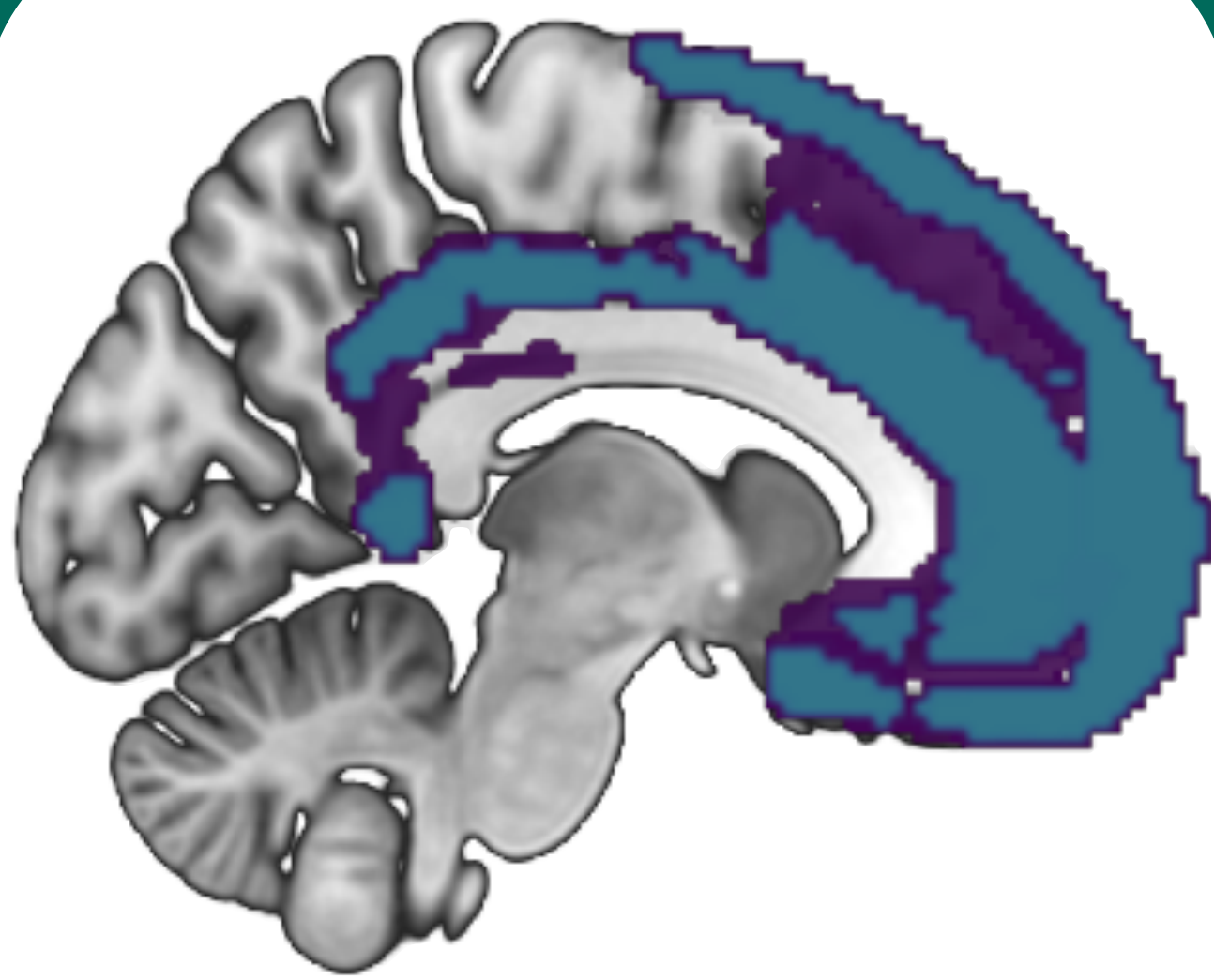
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## Rationale

**Pulsed arterial spin labeling (PASL)**<sup>1</sup> is an neuroimaging method that allows noninvasively estimating **cerebral blood flow (CBF)** in vivo. Prior studies<sup>2,3</sup> have found increased CBF in patients with lesions and neurological disorders, suggesting CBF could be a potential biomarker for these conditions. Mood disorders such as depression and anxiety often co-occur with epilepsy; however, few studies have investigated the relationship between the CBF and mood. It was hypothesized that participants with **idiopathic generalized epilepsy (IGE)** would exhibit greater mood disturbance than healthy controls (HCs), and that this would be associated with higher CBF in emotion-related brain regions. We also investigated whether seizure-free ( $\geq 6$  months) IGE participants (**IGE-**) differ in their mood status and emotion network CBF from those who still experience seizures (**IGE+**).

## Methods

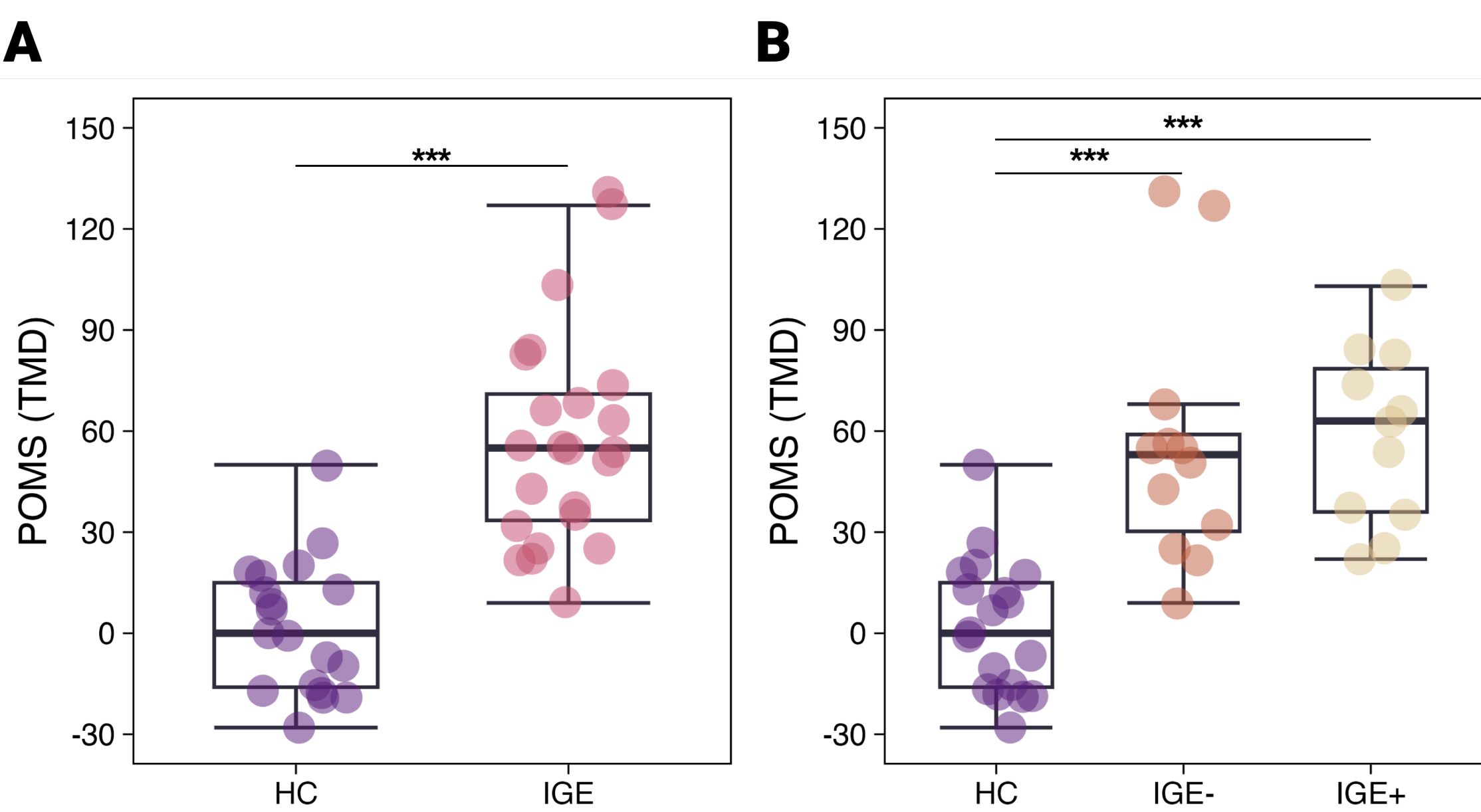
- 19 HCs, 23 IGEs (11 IGE+, 12 IGE-)
- Profile of Mood States (POMS)<sup>4</sup>
- **Total Mood Disturbance (TMD) = Negative** (tension, anxiety, depression, confusion, and fatigue) - **Positive** (vigor)
- Greater TMD  $\Rightarrow$  worse mood state
- Siemens Prisma 3T scanner with 20-channel head coil
- FSL package - BASIL
- PASL images corrected with M0 image  $\Rightarrow$  absolute CBF map
- Emotion Network (Fig. 01)
- Independent sample t-tests, Pearson correlation



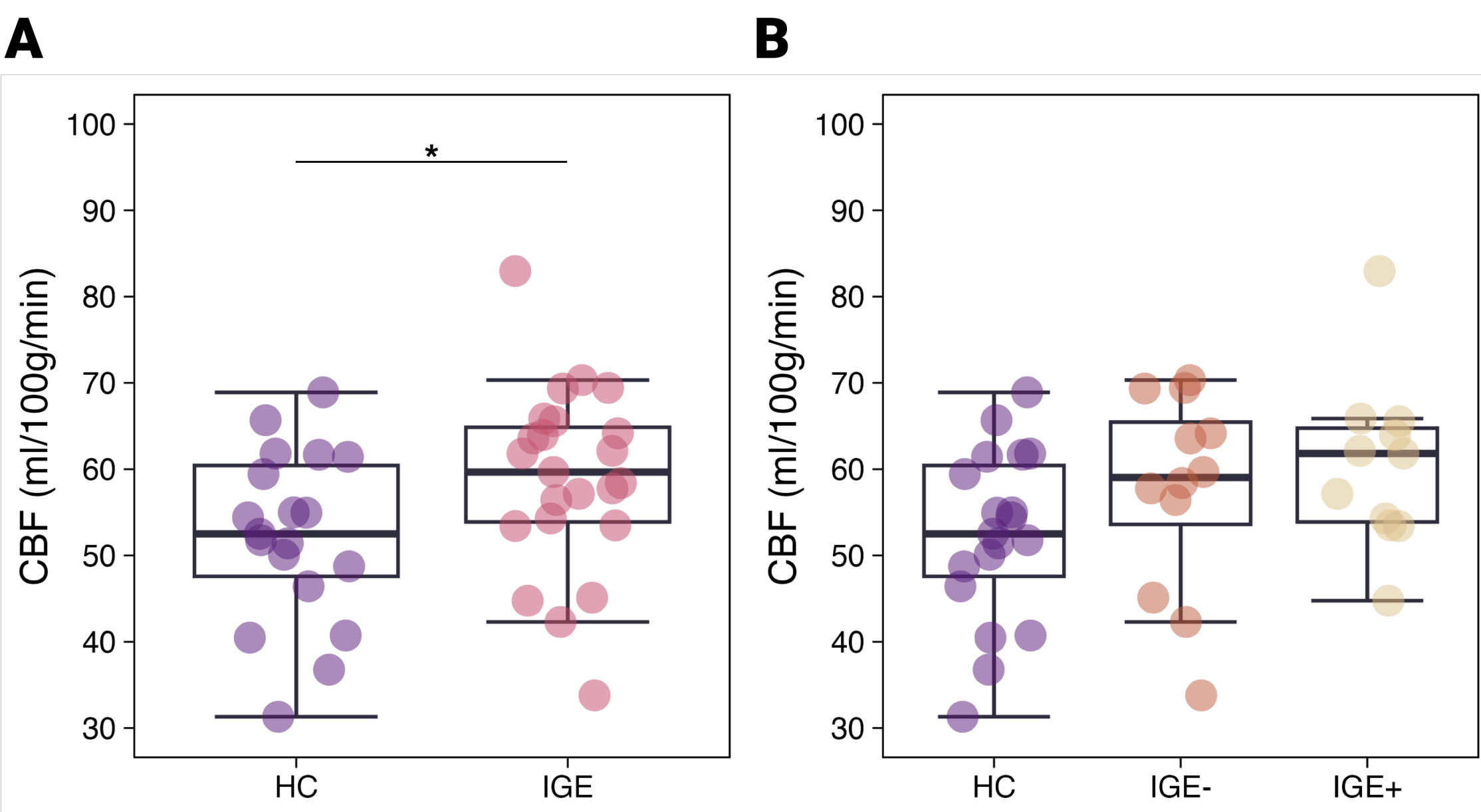
**Figure 1. Emotion Network**  
The mask contains bilateral prefrontal cortex, insula, hippocampal complex, and amygdala.

|                              | HC          | IGE         |             |             |
|------------------------------|-------------|-------------|-------------|-------------|
|                              |             | All         | IGE-        | IGE+        |
| N                            | 19          | 23          | 12          | 11          |
| Age, years                   | 29.5 (6.9)  | 31.3 (7.4)  | 31.2 (9.29) | 31.4 (5.03) |
| Sex, female                  | 9 (47.4%)   | 13 (56.5%)  | 8 (66.67%)  | 5 (45.45%)  |
| CBF (ml/100g/min)            | 52.3 (10.0) | 58.9 (10.7) | 57.5 (11.6) | 60.5 (9.9)  |
| Profiles of Mood States      |             |             |             |             |
| Tension-Anxiety              | 3.8 (6.4)   | 15.4 (7.1)  | 14.9 (6.7)  | 15.8 (7.8)  |
| Depression-Dejection         | 4.0 (3.6)   | 20.2 (12.6) | 18.5 (13.5) | 22.1 (11.8) |
| Anger-Hostility              | 5.0 (4.2)   | 13.0 (8.3)  | 14.2 (9.7)  | 11.8 (6.8)  |
| Vigor-Activity               | 18.5 (5.7)  | 12.0 (6.1)  | 13.6 (7.4)  | 10.4 (4.0)  |
| Fatigue-Inertia              | 6.4 (4.3)   | 11.9 (5.4)  | 11.2 (5.4)  | 12.7 (5.6)  |
| Confusion-Bewilderment       | 1.5 (3.2)   | 12.0 (5.3)  | 11.0 (6.1)  | 13.2 (4.2)  |
| Total Mood Disturbance (TMD) | 2.1 (19.9)  | 57.4 (32.3) | 56.2 (38.0) | 58.7 (26.5) |

**Table 1.** Demographic data and the statistical results for HC, IGE- and IGE+.



**Figure 2.** Difference in TMD between HC and IGE (A), and among HC, IGE- and IGE+ (B).



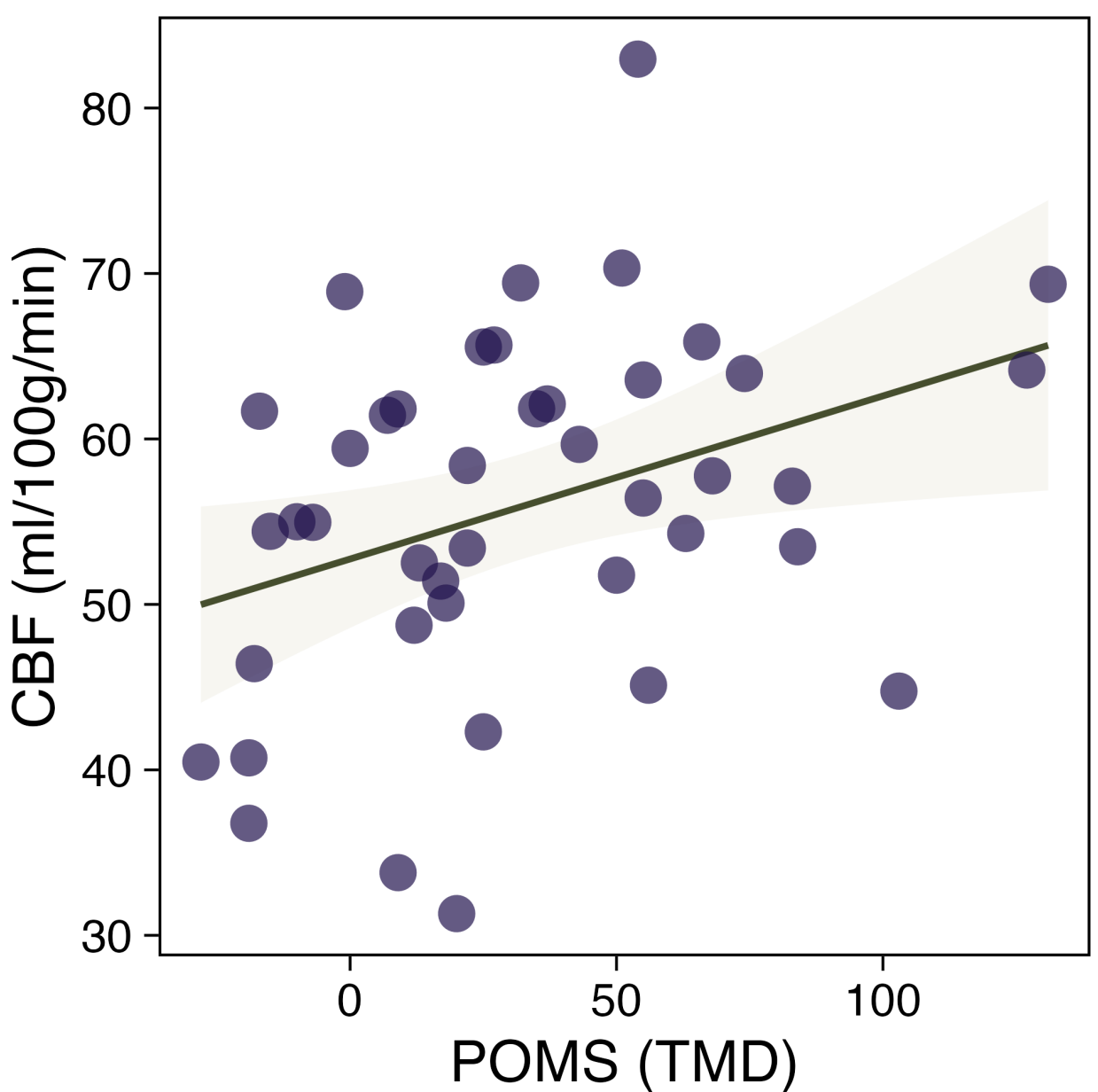
**Figure 3.** Difference in CBF between HC and IGE (A), and among HC, IGE- and IGE+ (B).

## Acknowledgments

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## Results

IGE had higher TMD than the HCs ( $p < 0.001$ ; Fig.01A); both IGE- ( $p < 0.001$ ) and IGE+ had higher TMD than HCs ( $p < 0.001$ ; Fig.01B) with no differences between IGE+ and IGE-. In regions comprising the emotion network, participants with IGE had significantly greater CBF than HCs ( $p = 0.044$ ; Fig. 03). Pearson correlation showed that TMD positively correlated with emotion network CBF ( $R = 0.36$ ,  $p = 0.021$ ; Fig. 04).



**Figure 4.** Correlation between CBF and TMD

## Discussion

- Other studies found hypoperfusion in epilepsy patients<sup>5</sup>, but our results demonstrated that CBF increases in epilepsy.
- The differences between IGE- and IGE+ were not significant and further investigations of the effects of seizure control on blood flow in mood networks are warranted.
- Future neuroimaging studies in larger samples and in other types of epilepsy, such as temporal lobe epilepsy, may be conducted to further investigate the relationship between mood state and CBF.

## Conclusion

Participants with IGE had significantly greater mood disturbance than HCs regardless of seizure status. Additionally, IGE participants had higher CBF (IGEs>HCs) within the emotion network, and also exhibited greater mood disturbance. These findings suggest that increased CBF within the emotion network may be associated with worse mood.

## References

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